

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Coding Challenge	Assessment:	Assessment Tool 2– Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	U.MIDHULA	Reg. No.:	192472115
Section / Batch:	CSE-AI	Date:	26-8-26

Code of Conduct

I 192472115(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U.MIDHULA

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to stand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Derive and implement Newton's Method (using the Hessian) for logistic regression optimization, and compare its convergence speed (number of iterations to converge) against standard Gradient Descent on the same dataset.

Code:

```
import numpy as np

X = np.array([
    [1, 1],
    [2, 1],
    [2, 2],
    [3, 2],
    [3, 3],
    [4, 3],
    [4, 4],
    [5, 4],
    [5, 5],
    [6, 5]
], dtype=float)

Y = np.array([0, 0, 1, 0, 1, 0, 1, 1, 0, 1], dtype=float)

X = np.c_[np.ones(len(X)), X]

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

theta_newton = np.zeros(X.shape[1])

for i in range(100):
    p = sigmoid(X @ theta_newton)
```

```

    gradient = X.T @ (p - Y)

W = p * (1 - p)

H = X.T @ (X * W[:, np.newaxis])

theta_newton = theta_newton - np.linalg.solve(H, gradient)

if np.linalg.norm(gradient) < 1e-6:
    break

newton_iterations = i + 1

theta_gd = np.zeros(X.shape[1])

learning_rate = 0.1

for i in range(10000):

    p = sigmoid(X @ theta_gd)

    gradient = X.T @ (p - Y) / len(Y)

    theta_gd = theta_gd - learning_rate * gradient

if np.linalg.norm(gradient) < 1e-6:
    break

gd_iterations = i + 1

print("Newton's Method")

print("Coefficients:", theta_newton)

print("Iterations:", newton_iterations)

print("\nGradient Descent")

print("Coefficients:", theta_gd)

print("Iterations:", gd_iterations)


print("\nComparison:")

```

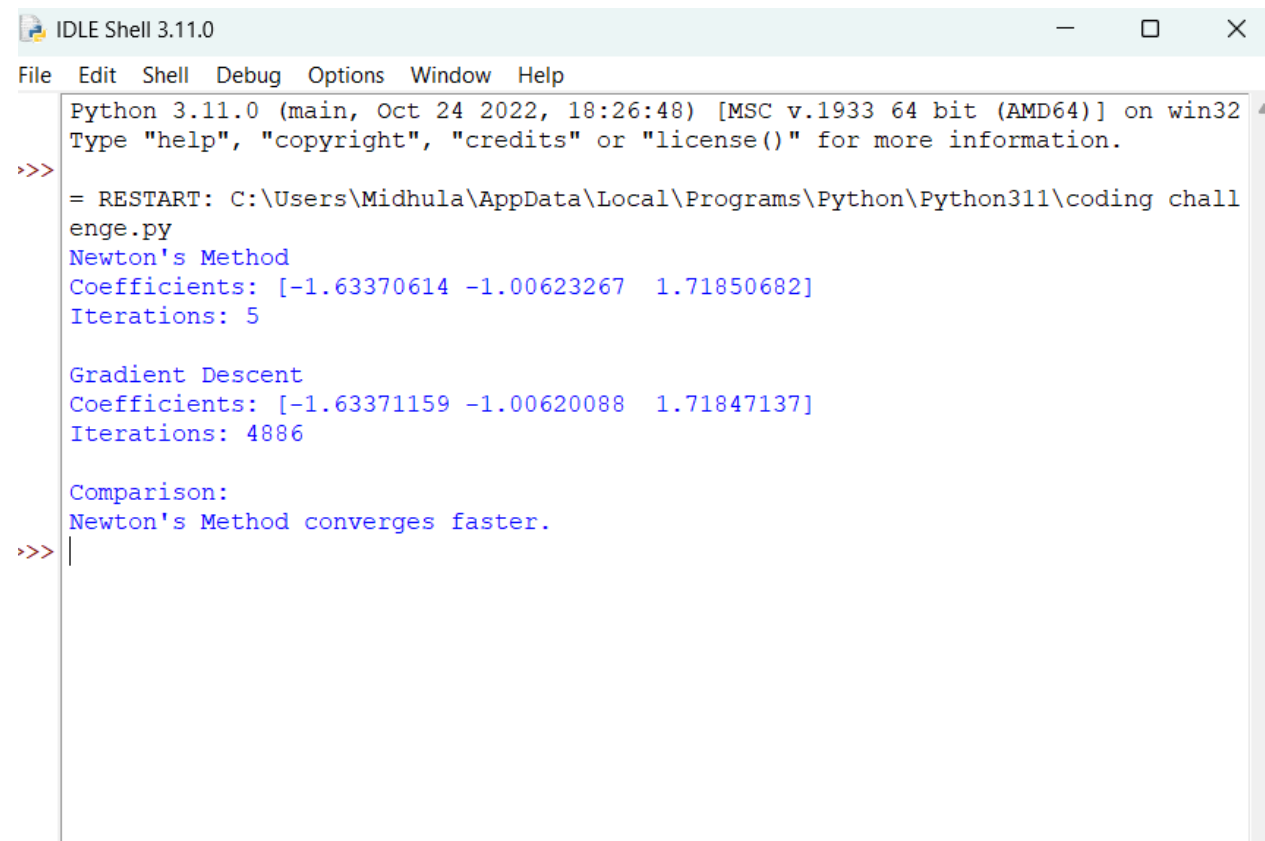
```
if newton_iterations < gd_iterations:

    print("Newton's Method converges faster.")

else:

    print("Gradient Descent converges faster.")
```

Output:

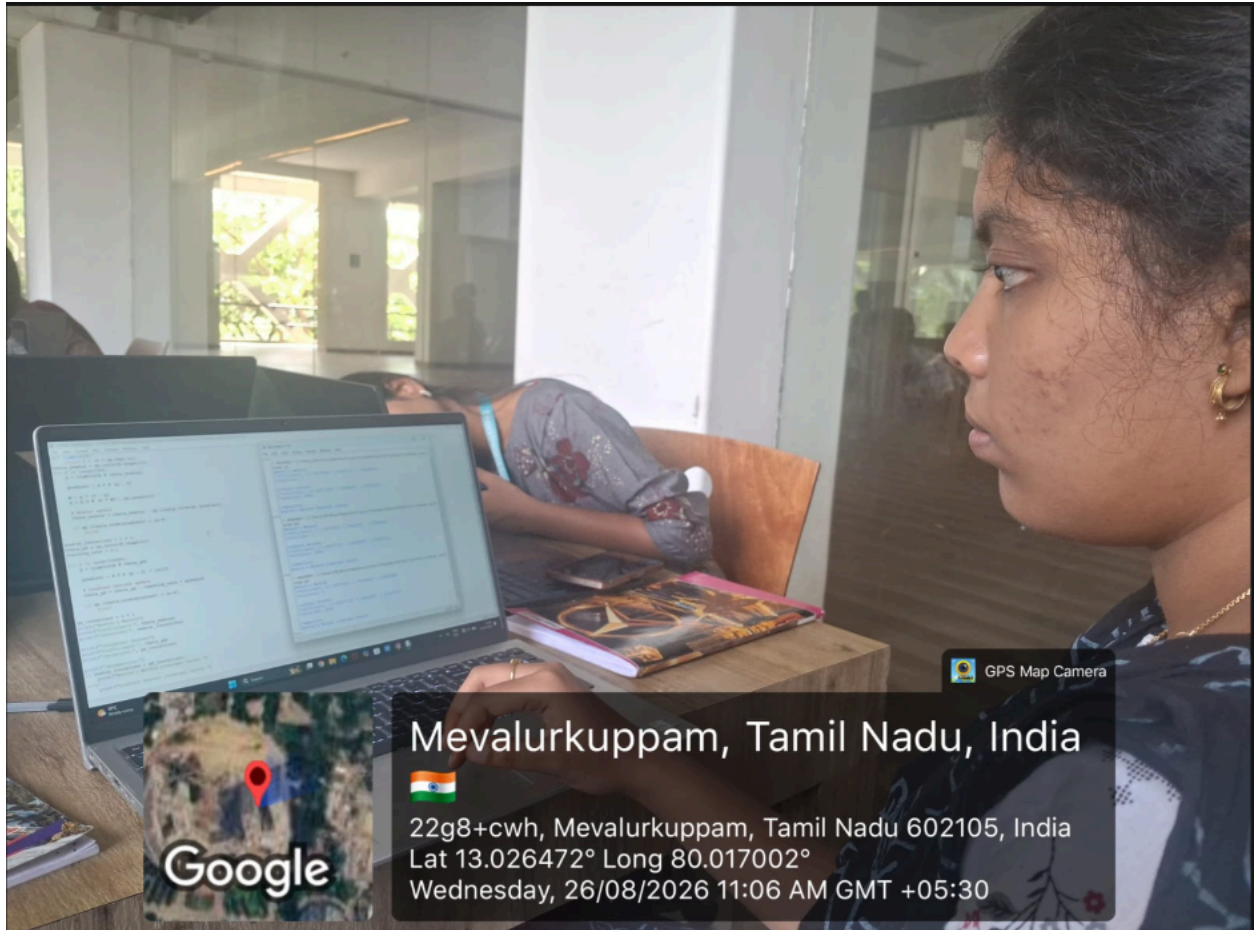


```
IDLE Shell 3.11.0
File Edit Shell Debug Options Window Help
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:\Users\Midhula\AppData\Local\Programs\Python\Python311\coding challenge.py
Newton's Method
Coefficients: [-1.63370614 -1.00623267  1.71850682]
Iterations: 5

Gradient Descent
Coefficients: [-1.63371159 -1.00620088  1.71847137]
Iterations: 4886

Comparison:
Newton's Method converges faster.
>>> |
```

Geo Tag:



GPS Map Camera

Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.026472° Long 80.017002°

Wednesday, 26/08/2026 11:06 AM GMT +05:30

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